

PRODUCT REQUIREMENTS DOCUMENT · SAMPLE ASSIGNMENT

Digital Heroes PRD *(Level 2)*

A custom web-to-print platform brief for full-stack development trainees. *Edition 2026*

Six print divisions, one owned platform. This document is the single source of truth for what you are asked to design, build, and ship during the selection process.

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ISSUED FOR SELECTION PROCESS ONLY

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What's *inside.*

A full read of the PRD assignment, from the brief to the build to how your submission will be judged. Sixteen sections, then a closing brief with everything you need to start.

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§ 01 · THE BRIEF

Project *overview.*

Build a custom web-to-print platform for a multi-division print business. Customers configure, design, proof, and order print products online; staff run every job from quote to production in one place.

PRD is a subscription-free, transactional web-to-print platform for a print house running six divisions under one roof: **labels & stickers, race & event numbers, MTB boards, stamps, trophies,** and **laser-cut work**. It must feel like a modern, Shopify-grade storefront, but do real production work the moment an order lands — not stop at an artwork upload.

Customers will be able to:

- Browse the six product ranges and see live examples of real print work.
- Configure a product (size, material, quantity) and get an **instant price**.
- Design artwork in-browser or upload finished files, then approve a proof.
- Pay, track the order, and **reorder in one click** from saved designs.

6	2	1
PRODUCT DIVISIONS	USER JOURNEYS	OWNED PLATFORM

§ 02 · CORE OBJECTIVES

Core *objectives.*

Pricing engine	Instant per-unit pricing by product, size and quantity, with automatic volume discounts.
Web-to-print designer	In-browser editing constrained to printable zones, with a real-time proof preview.
Flexible product logic	Parameter-driven products — dynamic sizing, not one SKU per dimension.
Print-file output	Automated high-resolution, production-ready files generated per order.
Accounts & reorder	Customer accounts with order history, saved designs and one-click reorder.
Production dashboard	An internal console to run every job through a clear pipeline.
Outstanding UI/UX	A confident, modern interface that does not look like a generic print store.

§ 03 · WHO USES IT

User *roles.*

Three roles share one backend. Define permissions clearly — what each role can see and do is part of the assessment.

Public visitor

- View the platform and product ranges
- Browse the live work gallery
- Use the instant price calculator
- Initiate signup / first order

Registered customer

- Manage profile & settings
- Configure, design & order products
- Approve proofs / request changes
- View order history & reorder
- Save and reuse past designs

Administrator

- Manage users & orders
- Run the production pipeline
- Download print-ready files
- Manage products & templates
- Access reports & analytics

§ 04 · PRODUCT CATALOGUE & CONFIGURATION

A catalogue *without bloat.*

A small, clean front-end catalogue powered by flexible logic, so the range can grow without flooding the admin with near-duplicate SKUs.

THE SIX DIVISIONS

LABELS & STICKERS · RACE & EVENT NUMBERS · MTB BOARDS · STAMPS · TROPHIES · LASER-CUT WORK

CONFIGURATION RULES

- **Parameter-driven products.** A single product models many variants — do not create one SKU per size.
- **Dynamic sizing.** Size, material, orientation and quantity are configured, not pre-listed.
- **Per-placeholder fitting rules.** Shrink-to-fit, tracking, and min/max font size so content always fits the print area.

> **Why it matters.**
> One flexible product replaces dozens of near-duplicate SKUs, cutting catalogue bloat and admin overhead in equal measure.

§ 05 · MONEY IN

Pricing & *checkout.*

Transactional, not subscription. The platform earns per order, priced live by an instant calculator — no quote forms, no waiting.

Pricing engine	Live per-unit pricing by product, size and quantity. Price updates instantly as options change.
Volume discounts	Applied automatically from a defined quantity floor (e.g. from 100 units).
Tax	VAT-inclusive totals shown at a glance; final price confirmed at checkout with no surprises.
Gateway	Stripe or an equivalent PCI-compliant provider via an integrated checkout.
Order capture	Payment and full order data written to the structured backend against the customer account.
Accounting-ready	Order records structured for clean export to accounting (Xero-ready data shape).

> **No subscription tax.**
> Revenue is per order. Do not build a recurring-billing model — build a clean, fast, per-order checkout instead.

§ 06 · WEB-TO-PRINT DESIGNER

Design inside *the lines.*

The designer lets customers build artwork in-browser, but only inside producible zones, so nothing unprintable reaches the production floor.

- **Template-driven editing.** Add and edit text, choose fonts and sizes, set colours, upload logos, and reposition, resize or rotate elements.
- **Print-area constraints.** Editing is locked to defined print areas; unprintable artwork cannot be created.
- **Real-time proof preview.** Artwork is shown in position — what is approved matches what is printed.
- **Upload path.** Customers may upload finished artwork instead (PDF, SVG, AI, PNG).
- **Saved designs.** Approved artwork is stored against the account for fast repeat ordering.

§ 07 · WHERE IT EARNS ITS KEEP

The print-file *engine.*

This is the part that separates a real web-to-print platform from a generic store: turning an approved design into a file a printer can actually run.

PRODUCTION OUTPUT

- **Automated print-ready files.** Per-order high-resolution PDF and PNG, with defined bleed and resolution per product type.
- **CSV variable-data merge.** Server-side processing generates one artwork file per row for race numbers and bulk jobs — no manual tweaking.
- **Per-placeholder fitting.** Shrink-to-fit, horizontal tracking and min/max font size so variable content always fits.

STRUCTURE BY DEFAULT

Outputs follow a predictable naming convention and are grouped in order-based directories:

`OrderID_ProductType_Date_Output`

> **Server-side where it counts.**
> Print-file generation and CSV merge must run on the server, keeping heavy processing off the customer's device and consistent across every order.

§ 08 · PROOF & VERSION CONTROL

Versioning that *holds up.*

Every proof revision is logged, so there is never a question of which file the customer actually approved.

- **Versioned proofs.** Each revision is stored as `v1, v2 ...` against the order.
- **Approval logs.** Record who approved which version and when.
- **Change requests.** Customers can request revisions before approving the final proof.
- **Approval gate.** Print-ready files unlock only once a proof is approved.

§ 09 · THE CONNECTIVE TISSUE

Accounts & *reorder.*

Customer accounts that remember everything turn a repeat job into a single click and keep proofs retrievable against the account.

- **Authentication.** Secure login and registration.
- **Order history.** Every past order, its status and its files, accessible to the customer.
- **Saved proofs & designs.** Approved artwork stored for reuse.
- **One-click reorder.** Re-place a previous order from saved artwork without redesigning.

§ 10 · CUSTOMER DASHBOARD

The customer *view.*

The signed-in customer dashboard must include all of the following modules.

- ✓ Account & profile status
- ✓ Order history with current status
- ✓ Score / configuration of saved designs
- ✓ Active proofs and their version state
- ✓ One-click reorder of past jobs
- ✓ Downloadable invoices / receipts

Note: the customer never sees production files or other customers' data. Keep the customer view and the admin view strictly separated by role.

§ 11 · ONE QUEUE, QUOTE TO DONE

Production *dashboard.*

The internal console is a single source of truth. Every order arrives with its artwork and configuration attached, and moves through a clear pipeline.

THE PIPELINE

QUOTE · ORDER · ARTWORK · PROOF · PRODUCTION · COMPLETED

REQUIRED CAPABILITIES

01 **Searchable job database.** Filter by status, customer, date and product type.

02 **Ready-for-production indicators.** Every order shows whether files are unlocked.

03 **Download production files.** Pull high-resolution PDFs and per-row CSV artwork from the order record.

04 **Advance the pipeline.** Move jobs through each stage with status updates.

05 **Reports & analytics.** Totals for orders, revenue and product-family throughput.

06 **Archive.** Completed work is retained for fast retrieval on repeat and event orders.

§ 12 · UI / UX REQUIREMENTS

Boutique *industrial.*

Clean, technical and fast — warm enough for makers and small businesses. It must not look like a generic, templated print store.

Feel

A confident accent colour driving every call to action, on a neutral base. Geometric headings, a body face built for dense specs on a phone.

Required surfaces

A product hub of equal-weight tiles as navigation, a live instant-price calculator, and a gallery of real work organised by product family.

Direction in one line: mobile-first, motion-enhanced, fast. Subtle transitions and micro-interactions — capable and modern without shouting over the products.

§ 13 · UNDER THE HOOD

Technical *requirements.*

A modern, transferable stack with nothing exotic. Heavy work runs server-side. The reference stack below is recommended; equivalents are acceptable if justified.

RECOMMENDED STACK

Frontend

NEXT.JS | REACT | TAILWIND

Backend

NODE.JS | SUPABASE | SERVER APIS

Storage & integrations

CLOUD STORAGE | PAYMENT GATEWAY | XERO-READY

NON-NEGOTIABLE

- **Mobile-first, fully responsive** across mobile and desktop.
- **Fast performance** — optimised assets, minimal blocking resources.
- **Secure authentication** — JWT or session-based, HTTPS enforced.
- **Email notifications** — order updates, proof status, completion alerts.
- **Server-side file generation** for print files and CSV merge.

§ 14 · SCALABILITY CONSIDERATIONS

Built to *grow.*

- Architecture must support adding new product divisions and template groups without rework.
- Data structures extensible toward corporate / multi-account ordering.
- Codebase structured to support a future mobile-app version.
- An accounting-integration path (full Xero API) ready to switch on later.

§ 15 · WHAT YOU HAND OVER

Mandatory *deliverables.*

Live website	A fully deployed, publicly accessible URL.
Customer panel	Test credentials; signup / login / configure / design / proof / dashboard all functional.
Admin panel	Admin credentials; user management, production pipeline, file download, reports.
Database	Backend connected (e.g. Supabase) with a proper, documented schema.
Source code	Clean, structured, well-commented codebase.

DEPLOYMENT CONSTRAINTS

- Deploy to a **new** Vercel account — not a personal or existing one.
- Use a **new** Supabase project — not a personal or existing one.
- Environment variables must be properly configured and never committed.

§ 16 · EVALUATION & TESTING

How it's *scored.*

Requirements interpretation	How accurately requirements are translated into working features.
System design	Quality of architecture decisions and data modelling.
UI/UX creativity	Originality, polish and engagement of the interface.
Data handling	Accuracy of pricing, file output and order logic.
Scalability thinking	Extensibility of the codebase and data structures.
Problem-solving	How ambiguous requirements are identified and resolved.

TESTING CHECKLIST

- ✓ Signup & login
- ✓ Product configuration & live pricing
- ✓ Volume-discount calculation
- ✓ Designer — print-area constraints
- ✓ Proof versioning & approval gate
- ✓ Print-file & CSV merge output
- ✓ Reorder flow
- ✓ Admin pipeline & file download
- ✓ Responsive on mobile & desktop
- ✓ Error handling & edge cases

Build the *platform.*

This brief is your single source of truth. Where it is silent, make a reasonable decision, document it, and be ready to defend it — identifying and resolving ambiguity is part of what we are assessing.

DOCUMENT
PRD — Sample

PURPOSE
Trainee selection assignment

VERSION
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AUDIENCE
Full-stack development trainees

WEBSITE
digitalheroes.co.in

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